

* SAVE POINT (To set a point within a transaction to which you can roll back)
purpose: manage transaction within a database.

String is called in linux in

Varchar

* where - it is select from the place like column, row.

* desc - it is describe the table.

* order by - to order the values in ascending and descending.

*) Constraints - Set by rule in column

*) primary key - it is allowed to unique values and it's not allowed to duplicate values and NULL

*) Composite primary key - it is also called primary key (But constraints two (or) more primary key are set to the table.)

*) Unique key - It is also work same to primary key but it's allow to NULL.

* Surrogate key - it's automatic create a primary key like (S No)
this automatic key is called surrogate key.

* Candidate key - all primary key and composite key is called candidate key. it is not allowed to optional

* Super key - it is a concept of it is also candidate key is super key.

→ aggregation functions

It is the function of (minimum, maximum, sum, average, Count) and ~~etc~~ it's called aggregate function.

* MIN

* MAX

* SUM

* COUNT

* AVG

* Having - it is worked to where because where not worked is the function.

String handling

- *) Length - length is convert the length to the string.
- *) Upper - the string is convert to the capital letter
- *) Lower - the string is convert to the small letters.
- *) Concat - this function are joined to the strings
- *) Substring - to get the large count string to cut the important one.

ex:

Ravi kumar. to get 6 count

to get Ravi / k.

- *) Trim - to remove the additional spaces in the string.

to remove Right and Left in string space Use RTrim and LTrim

* Case Condition
The Case Condition is the function
of these,
* if
* else, this is case condition
Use - when.

Null handling

Null is using actual NULL not 'null'
this is string null. actual null
appears to get the value use
= is NULL. but to get string
null use - = NULL.

this format is accept
the Not Null in actual and
string.

* ~~Coalesce~~ Coalesce - this function
use reduce to the null and
it's not null accept.

* Left Join

Left join is add two table this give common value and left side table value, but right side table value in left side table value is None.

* Right Join

it is same to left join work this give all right side value and not give left side value it's None

* Self Join

Self join is join in same table it's called self join.

Window Function

The window function in SQL performs the calculation a set of rows.

without changing the number of rows in the result set.

Types

- * Aggregate
- * Ranking
- * Value.

, *lpad - R pad) - this function
to get Right and left is
counts but actual will give
to counts in the string to
create is can't to add (+)

*) Replace - replace the value in
strings one value like
space remove to underscore add

*) Instr - this function to get
the count from the letter
position.

ex: priya (to get) a.

position 5.

*) Reverse - the string are convert
to the reverse form.

*) Format - the value to
convert to formatly
like, 410, 345.00. this is
one format.

Analyze

this function analyze the table time and cost analyze to give.

partition

partition is divided to the space.

this partition is search any data's in the partition find then it's fast and best performance.

these are three types in mysql

* Range partition (i)

* List partition (ii)

* Hash partition (iii)

(i) Range partition
partition by range.

this partition is divide in the partition in year. (range)

ex : 2020 and 2021, the data

save in 2020 data are save 2020 and 2021 data are stored 2021 partition.

(ii) List partition

this is divided to the partition in list like (Ho, manager, TL, member)

the data's are stored in the type of employee in the partition.

Aggregate

* AVG()
* MAX()
* MIN()
* SUM()
* COUNT()

Ranking

* ROW_NUMBER()
* RANK()
* DENSE_RANK()
* PERCENT_RANK()
* NTILE()

Value

* LAG()
* LEAD()
* FIRST()
* LAST()
* NTH_Value()

Union and Union all table.

the Union all function is add the table it's work with Full join this adds row's is same no of column's and datatype comparable.

the Union function is also Union all function, but remove the deduplication, in the data's.

Index Concept

the Index Concept is optimize the performance in the SQL.

it's shortly work from the index method, the data's are stored to index and get the best performance in Finding, etc

Explain

this function are explain the table in the data's

ex: Partition name, index working or not working to give

Hive

Face book developed from the Hive
it's one of the platform in map reduce
it's a query engine to query SQL it's
convert to the map reduce.

Function of Hive

- *) show databases ;
- *) show tables ;
- *) create database name ;
- *) use name ;
- *) Load
 - *) Load data Local inpath - File
Load from Local device.
 - *) Load data inpath - File Load
from other data base.
- *) Insert - add data to
the loaded table.
- *) Hive internal tables and external tables.

partitioning

* Static partition

* Dynamic partition.

partition are created to real time
in time stamp like (year - month - Date)

Grant and revoke

this function are admin function
the newbie are join the job to
create the user id to give this
grant.

(or) the employee are resigned
the job to revoke (cancel)
the id this is revoke.

Python JDBC Connectivity

to install package in python terminal
to install package (pip install mysql-
connector-python) in terminal

(or) using Pycharm to download
and install package

Normalization

Normalization in sql is a systematic
process of organization the column and tables
(or) attributes and relations.

See about the three type of
Normal Forms

* 1 NF

* 2 NF

* 3 NF

(iii) Hash partition

Hash partition are divided in not find to the list are range the other type partition is called Hash partition

Date & Time

date time and time stamp it's work is save to the event time. but there are two type is different from back ground.

Regular expression — regex

it is handling the string in sql. it is help to get data in data base (or) table.

it's used to cheat words. the cheat words are get gitup in Data engineering channel.

Commit

Commit is the function of the save the function this is commit
Roll back

to check the value correct this function one time give commit then not give the roll back function.

1 NF

1 NF requires that every column holds only atomic values and that there are no repeating groups.

2 NF

2 NF requires that table to be in 1 NF and that all non-key columns are fully functionally dependent on the entire primary key.

3 NF

3 NF requires that table to be in 2 NF and that all columns are directly dependent on the primary key (i.e. transitive dependency).

SCD (Slowly Changing Dimension)

is a method used in databases to track and preserve historical changes in data over time.

There are three types.

- * Over write (delete and add)
- * Add new record (continue add)
- * Add new attribute (add new column).

ACID - Atomicity, Consistency, Isolation, Durability.

Atomicity

it is the function of all
is run pass it's pass but one
fail it all function fail it is
Atomicity.

Consistency

Data is any where is valid
it is Consistency.

Isolation

two process are run at a time
it's run in don't crash the
function in the table.

Durability

it is the Commit the value
but crash any one side but
are run in recover to the
page to confirmation in online.

Spark

The prerequisite is known any programming language like (Java, Python, scala) and SQL.

the spark is using Data processing, and parallel (or) distribute processing.

Spark Frameworks

- * Spark Core
- * Spark SQL
- * Spark Streaming
- * MLlib (Machine Learning Library)
- * GraphX

MLlib is for fully data science

Field work

GraphX is processing the graph based (or) mutual data processing. its graph data stored.

The data engineers are mostly used the spark core, spark SQL, spark streaming.

Spark RDD

Resilient distributed dataset
this is the fault tolerant in the replication (or) recompute the lineage data in distributing

RDD → DataFrame → dataset.

spark transformation and action

ala) RDD work is to divide to sub. types this is the function transformation and action.

transformation:

first program to read, map operation and flat map this type

function is called transformation.

transformation is narrow and wide two types.

action:

the program to collect and show etc function is the types is action.

→ spark is run in lazy evaluation it is one action is called whole program is running to action program it not worked.

spark executor and memory

it is the program property of the spark.

* num - executor (1)

* executor - cores (2)

* executor memory (3)

* driver memory (4)

Bucket

→ distribute the data like partitioning.

* the tables are data's are partitioning by only run mapping function ~~one~~ spark. this process run only create partitioning and load data.

* First create non partition table to load the data on the table and then to insert partition table.

→ Buckets are used in the hash partition algorithm.

→ it's used to sampling.

How to find bucket count:

① table size = ex: 2300 mb

② Block size in HDFS in the table = ex: 128 mb

$$2300 / 128 = 17.96$$

Formula:

$$2^n = 17.96$$

exell
 $N = \lceil \log_2(17.96) \rceil$

$$2^5 = 32 \rightarrow \text{bucket count}$$

Hive partition with bucket

get data's from the user one partition are created but that data are not create the sub partition then to create the bucket.

Spark Lense

It is the profiling tool in spark. this is calculate in system to show the executor, memory, and tasks how many this calculation tool.

Hash partition in spark

this is the function of the automatic build partition in the spark this is also same as map reduce function.

algorithm:

= $\text{hash of the key (mod)} \% \text{ no of o/p task count}$

Custom partition in spark

this custom the data's store the data's location to set the partitions to create.

Repartition and Coalesce

the repartition is the increase of the partition and coalesce is the decrease the partition of the spark. it two function common in shuffling repartition is full shuffle but coalesce is decrease 2 that one partition

example

memory

6 nodes and each node have 15 cores, 64 GB RAM
Cores - no of concurrent tasks

For example we can give 5 parallel tasks for
each executor

num cores : 5

num executor : $15 / 5 \Rightarrow 3$ executor per node.

num executor memory : $64 / 3 \Rightarrow 21$ GB ~~10~~ 2
GB for yarn memory = 18 GB (we can
have 5 GB to 20 GB memory for each job)

driver memory : the driver memory is fixed
to the same to executor memory.

spark submit

recommended to eclipse to use the
program build to scala and submit
the program. it's convert the jar file

Run command for terminal (spark submit)

[bin / spark-submit --class com. spark-example.
wordcount --master yarn --deploy-mode cluster
--num-executors 1 /home/big /work space -3/
spark-example /target /spark-example-0.0.1-
snapshot jar -with-dependencies ~~with~~ . jar]

File is 70% File 1 partition that
next 30% 2 partition. the repartition and
coalesce are increase and decrease the
partitions but none fit is see about
repartition - increase, coalesce - decrease use.

Spark dedup

it's remove the duplicate records and
files to delete it's called drop duplicate
(or) dedup.

UDF Function

The user defined function is also known
as the hive udf function. This function
is created the user.

reduce by key and group by key

this is the wide transformation in
spark. reduce by key is give the best
performance and group by key is comparatively
low performance in spark.

reduce by key decrease the shuffling
then performance is high.

Moving average

moving average is the calculation to analyze
the data points to create the series of avg
the data sets in moving averages.

ls -ltr - give a detailed information to list.
ls -a = to list a hidden file

ln - hardlink create (single data (or) file to get a two names) or create two names

rm - remove the file, document, data, file

ln -s - create softlink (it's copy the path of file in original file) it's original
Delete all data and file)

nohup filename - the process are run in the background

nohup filename >> filename &
→ Log file name

top - task manager for background

ps -aux - it's also task manager. For background only

kill -9 process_id - to kill the background process.

df -h = to show the files and storage.

du -sh = to show the single file usage.

tar -czvf = Compress the file

tar -zxvf = extract the file.

free -g = to see the ram value.

sudo sh -c "sync; echo 3 > /proc/sys/vm/drop_caches" = to remove the buffers and caches.

Sqoop

SqL + Hadoop this is join the technology
is sqoop.

to transfer the data to RDBMS
to Hadoop.

Sqoop is not work to reduce only
work on map function.

(Linux)

PWD - present working directory

Whoami - give a user name

Uname -a the system bit type (like 64
32) and OS name.

mkdir - make a directory

CD - change directory

Sudo apt-get install appname - install application.

touch - create a new document in empty

cat filename - to view the file in terminal

echo - is the print to the terminal
in linux.

~/ - it's home file short cut

mkdir -p - it's create directory inside

CD .. - it's create another directory
backward to last directory outside

AWK - it's also cat function to view
seperate column and row.

wc - this command to count the line, words, and characters (letters)

sort - to view the alphabetic order on the file or document values

sort -r = is to sort the value reverse

sort -n = it's order the numbers

head - it's show the first 10 lines

tail -f - it's show the last 10 lines

grep - to find the file and word to the system.

SSH - to connect the app putty app to connect the other device. IP address

scp - to copy to the other device to IP address

find - to find the files, documents and etc.

→ find command to find the files, directories and it's locating the by name, size, type and etc
(example - find . -name "*.log")

→ grep command to find the inside files and texts it's searching like lines/matching

example - grep "Error" log.txt

1) Num executor

The numbers of executor are fixed in the nodes like 10 executor this is num executor.

3) executor code

The executor are run in the task to save the memory to run task this memory is called executor memory.

2) executor memory

The single node are work in parallel work in the (5 to 6 (any num)) to take care spark task it's parallelly work in every node the task to create this is executor core.

6) Driver memory

Code main program is run in any node in, one collect and show related function is stored outputs in driver memory the tasks output. it's sometimes only is used. this is driver memory.

spark to set memory properties

it is also known the the executor driver and spark memory RDL to the calculations.

Big Data — Data Engineering

big data providers.

Hadoop

it's divide to two types

- * HDFS - Hadoop Distributed File System
- * MR - Map reduce.

HDFS

it's derived from the google file system. it's distributed the blocks.

Map reduce

it's derived from the google map reduce. it's connected to the distributed blocks.

Quota

Quota are divided in two types.

- * space quota
- * set quota

space quota are fixed the input data size like (100 bits) Less than 100 bits are save the data

set quota are fixed the file no any size data are inputted but like (3 no) only put the 3 files.

Sub query's

Sub query is one query in another query is called sub query.

ex:

Select (Select), etc.

View

View is simplified to the my sql code and decrease the code (query) to connect big command in view to simply name run to execute the table.

Joints

Joints is help to data engineer work and time management. because process are run several tables in one time then add to the table joint so give the answer

It is easy process and work efficiency.

Joints are several types:

* Inner Join

this join two table and get the value for common value in the table.

- * Not Null - table are not run to the value of place in Not null that's are not work the process
- * check - insert the condition in the table.
- * Foreign key - it's create a relation from two table
- * default - it is the value to assign the value in to create the table.
- * Not minimal - it is the composite primary key used. it's work on one remove but work on the uniqueness this is (Not minimal)
- * minimal - it's not removed the one column it's not working this is minimal.
- * Natural key - this is naturally unique in the value like (Aadhar, phone no). it's also called and used (primary key). it's identify the category.

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Previous Tasks

1) Create two variable assign any values that should be store in third variable and the output will be in the "Count"

2) Get a input for first name and last name output will be add the full name

3) Create a variable of Name, age, height and finally use boolean operation to ask you

are student the output will say yes/no.